



Deep Learning Course

Course Faculties



- PhD in System Analysis
- Head of DS at [Prog.AI](#)
- Lead Engineer at Dun & Bradstreet
- Lecturer at KSE and IASA KPI
- Kaggle Expert and winner of lots of Ukrainian Kaggle Competitions



- PhD in System Analysis
- AI Product Lead in Better Medicine
- Co-Founder/Head of AI at Zvook
- Machine Learning advisor at Mantis Analytics
- Kaggle Competition Grandmaster. Top 150 World Competition Rating
- Lecturer at KSE, IASA KPI, and UCU

Course Faculties



- Artificial Intelligence student at KSE
- Multiple AI/ML hackathon finalist and prize winner
- Kaggle Enjoyer



- AI/ML Engineer at WINWIN AI CoE
- Researcher in KSE NLP Lab
- AI27 Student
- TA at KSE CSD

Course Modules



- Module 1 – Deep Learning Basics
 - Lecture 1 – Deep Learning Overview, PyTorch, Perceptron (Volodymyr)
 - Lecture 2 – Backpropagation & Advanced Optimization (Volodymyr)
 - Lecture 3 – Validation & Metrics (Anton)
 - Lecture 4 – Regularization: Dropout, L1/L2, Batch/Layer Normalization (Volodymyr)
 - Lecture 5 – Frameworks & Logging: PyTorch Lightning & Weights & Biases (Anton)
- Module 2 – Convolutional Neural Networks (CNNs)
 - Lecture 1 – Intro to Image Processing, Convolutional & Pooling Layers (Volodymyr)
 - Lecture 2 – ResNets & EfficientNets (Volodymyr)
 - Lecture 3 – CNNs applications for Audio ML (Volodymyr)

Course Modules



- Module 3 – Recurrent Neural Networks (RNNs)
 - Lecture 1 – Intro to Sequence Tasks, Representations & Embeddings (Anton)
 - Lecture 2 – RNNs (Anton)
 - Lecture 3 – GRUs & LSTMs (Anton)
- Module 4 – Transformers
 - Lecture 1 – Attention from Scratch, Encoder Transformer (Anton)
 - Lecture 2 – Encoder–Decoder Transformer (Anton)
 - Lecture 3 – Transformers Beyond Text: (Anton)
 - Lecture 4 – Kaggle “Black Magic” Fine-Tuning & Distillation (Volodymyr)

Course structure and Rules



- 4 Modules:
 - 3-5 Lectures per module.
 - One BIGGGG Practical task per module.
- Lab Assignments (4x15 points):
 - Google form to submit Practical task.
 - The deadline is HARD!!! If you do not fit the deadline, you will get 0.
 - Maximum score = 15.
 - You can try to defend your assignment several times **till the deadline**
 - One student (group of students) can submit only one Google form.
 - Up to 3 students in a group
 - If you experience any problems, contact lectors directly.
 - Defence of each assignment will be **completely offline - on practices**
- Final Exam(30 points)
 - Individual Offline exam + defenses of the answers
 - Topics: All course material. Theoretical questions + proofs and practical tasks
 - Maximum score = 30

Course structure and Rules



- Midterm Test (5 points):
 - Individual offline small test for one practical and one theory question
- Preliminaries test (5 points)
 - Type: Individual written assignment to evaluate the preliminary knowledge of the student.
 - Grading: 5 points.
 - Timeline: During the 1st practice session

Tools



- Communication in Slack
- Lecture materials on GitHub
- Something could be happening in Moodle
- Homework can be submitted through:
 - GitHub;
 - Kaggle Notebook;
 - Google Colab.

Evaluation and Outcome



- The main outcomes are:
 - Make you interested in Deep Learning.
 - Give you a new knowledge.
 - Build a community.
- Evaluation:
 - Check all evaluation criteria in Syllabus
 - You can get additional points by:
 - Participating in Kaggle competitions. **BUT discuss your participation in any competition with lecturers in advance.**
 - Accepted Journal/Conference Papers. **Again discuss Journal/Conference and paper topic with lecturers in advance.**
 - Implementing existing papers (only if such an implementation does not exist). **Again discuss paper to implement with lecturers in advance.**